

SLC4A1 mutations that cause distal renal tubular acidosis alter cytoplasmic pH and cellular autophagy

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eLife Assessment

This work characterizes the function and localization of SLC4A1 variants associated with distal renal tubular acidosis in human patients. Cell culture and limited animal studies provide partial but **incomplete** support to the authors' claim that the variants disrupt normal protein degradative flux by alkalinizing the intracellular pH. The study is **valuable** in providing preliminary evidence for future exploration of the link between intracellular pH regulation by SLC4A1 and kidney cell function in vivo.

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Abstract

Distal renal tubular acidosis (dRTA) is a disorder characterized by the inability of the collecting duct system to secrete acids during metabolic acidosis. The pathophysiology of dominant or recessive *SLC4A1* variant related dRTA has been linked with the mis trafficking defect of mutant kAE1 protein. However, *in vivo* studies in kAE1 R607H dRTA mice and humans have revealed a complex pathophysiology implicating a loss of kAE1-expressing intercalated cells and intracellular relocation of the H⁺-ATPase in the remaining type-A intercalated cells. These cells also displayed accumulation of ubiquitin and p62 autophagy markers. The highly active transport properties of collecting duct cells require the maintenance of cellular energy and homeostasis, a process dependent on intracellular pH. Therefore, we hypothesized that the expression of dRTA variants affect intracellular pH and autophagy pathways. In this study, we report the characterization of newly identified dRTA variants and provide evidence of abnormal autophagy and degradative pathways in mouse inner medullary collecting duct cells and kidneys from mice expressing kAE1 R607H dRTA

mutant protein. We show that reduced transport activity of the kAE1 variants correlated with increased cytosolic pH, reduced ATP synthesis, attenuated downstream autophagic pathways pertaining to the fusion of autophagosomes and lysosomes and/or lysosomal degradative activity. Our study elucidated a close relationship between the expression of defective kAE1 proteins, reduced mitochondrial activity and decreased autophagy and protein degradative flux.

Introduction

Distal renal tubular acidosis (dRTA) is a disorder characterized by the inability of the collecting duct system to secrete acids during metabolic acidosis¹. In addition to hyperchloremic metabolic acidosis, patients with this disease can present with hypokalemia, kidney stones, urinary sodium waste and difficulty to thrive. Expression of pathogenic variants in the *ATP6V0A4*, *ATP6V1B1*, *FOXI1*, *WDR72* and *SLC4A1* genes are the usual genetic aetiologies^{2–4}. The *SLC4A1* gene encodes the anion exchanger 1 (AE1) protein, which is an electroneutral chloride/bicarbonate exchanger⁵. It exists in two forms: a 911 amino acid erythroid isoform known to interact with erythroid cytoskeletal proteins and participate in red cell respiration and integrity, and a 65 amino acid (NH2-terminal) truncated isoform primarily found in the basolateral membrane of renal type A intercalated cells (A-IC)^{6,7} and podocytes⁸. This isoform participates in bicarbonate reabsorption and through its physical and functional interaction with the cytosolic carbonic anhydrase II and apical H⁺-ATPase, it supports apical proton export and urine acidification⁹.

Pathogenic variants in the *SLC4A1* gene can result in either red cell defects (such as Southeast Asian ovalocytosis^{10,11} and hereditary spherocytosis¹²), renal cell defects (dRTA¹⁰) or both in patients with homozygous (Band 3 Coimbra and Band 3 Courcouronnes^{13,14}) or compound heterozygous variants^{15,16}. Renal *SLC4A1* disease-causing variants have only been found in the transmembrane domain - where it could impact protein structure and its transport function- or in the short carboxyl domain - where it possibly affects protein-protein interactions. The pathophysiology of dominant or recessive *SLC4A1* related dRTA (hereafter named dRTA) has originally been linked with the mis-trafficking defect of mutant kAE1 protein^{10,17}. However, recent *in vivo* studies in mice and humans have revealed a complex pathophysiology where a loss of kAE1-expressing intercalated cells, the intracellular relocation of the H⁺-ATPase and accumulation of autophagy marker p62 and ubiquitin-positive material in the remaining type-A intercalated cells was suggested to be the principal cause of the disease¹⁸.

The highly active transport properties of collecting duct cells require the maintenance of cellular energy and homeostasis. The autophagy-mediated turnover of damaged organelles is necessary for protecting collecting duct cells as in most renal cells¹⁹. The chloride/bicarbonate exchange function of kAE1 in A-ICs confers a pivotal role in pH homeostasis and thus is a major contributor to cellular homeostasis. kAE1 protein interacts with several proteins such as integrin-like kinase (ILK), adaptor-related protein complex 1, 3 & 4 (AP-1, AP-3 & AP-4 mu1A), transmembrane protein 139 (TMEM139), kinesin family member 3B (KIF3B), and clathrin among others^{7,20–22} that support protein stability and trafficking. It also interacts with homeostatic proteins including the glycolytic enzyme glyceraldehyde-3-phosphate dehydrogenase (GAPDH)²³ and the antioxidant enzyme peroxiredoxin 6 (PRDX 6)²⁴, which play major roles in cellular energy metabolism and oxidative stress response, respectively.

In this study, we report the characterization of newly identified dRTA variations and provide evidence of abnormal autophagy and degradative pathways in cells and kidneys from mice expressing dRTA mutant kAE1 proteins.

Concise Methods

Ethics Approval

This study was conducted in accordance with all national and institutional animal care guidelines and approved by the University of Alberta's Animal Care and Use Committee (AUP #1277).

Newly identified *SLC4A1* variations from dRTA patients

The patient carrying the R295H mutation was a boy carrying the variation in the homozygous state, whose genetic diagnosis was made at the age of 5, following growth retardation of -2 SD for both weight and height, with bicarbonate at 18 mmol/L, potassium at 2.8 mmol/L, chloridemia at 94 mmol/L, calcemia at 2.55 mmol/L, and a urinary pH of 7.5. He also had a history of a pyeloureteral junction syndrome that was surgically managed at the age of 3. The R295H dRTA variation is a nonsense homozygous substitution characterized by a replacement of guanine (G) on position 884 by adenine (A) in the coding sequence. It has an allelic frequency of 0.14 % in the European population.

For the Y413H variant, the patient was a female diagnosed at 1 month old, with an urinary pH of 7.5, evidence of nephrocalcinosis and failure to thrive. The S525F variant has been previously reported²⁵, but in brief, the patient was a 13 year old female with plasma pH of 7.25, plasma bicarbonate at 15.3 mmol/L who also presented with polyurethral junction syndrome, nephrocalcinosis and nephrolithiasis since childhood. The Y413H and S525F dRTA variations are nonsense heterozygous substitutions characterized by a replacement of thymine (T) at position 1237 by cytosine (C) and a replacement of cytosine (C) at position 1574 by thymine (T), respectively. The R589H dRTA variation has been previously described¹⁸. No follow up data were available for all patients.

Mice

Transgenic mice carrying the R607H knockin (KI) mutation (murine equivalent to human R589H mutation) were previously described¹⁸. Homozygous mice used throughout the study display incomplete dRTA with alkaline urine without metabolic acidosis at baseline as previously reported¹⁸. Homozygous mice or wild-type (WT) littermates were fed a standard rodent chow (Picolab® Rodent Diet 20 # 5053, LabDiet, ST. Louis, MO, USA) or for **Figure 5** (H-K), a salt-depleted diet with acid challenge as previously reported²⁶ with adequate and constant water supply, and maintained on a 12-hour light and dark cycle throughout their lifespan.

Statistical analysis

All the experiments were independently repeated a minimum of three times. Experimental results were analyzed using the GraphPad Prism software and are summarized as mean ± SEM. All statistical comparisons were made using unpaired Student's t test or one/two-way ANOVA followed by a post-hoc test as indicated in figure legends. A P-value less than 0.05 was considered statistically significant. All datasets were assessed for normality, and outliers identified by Prism were excluded.

Detailed Material and Methods are listed in the Supplementary Material

Results

The dRTA kAE1 variants traffic to the basolateral membrane but have reduced transport activity in mIMCD3 cells

We first characterized three newly identified dRTA mutations and compared them with kAE1 WT or previously characterized kAE1 R589H mutant. **Figure 1A** depicts the alpha fold predicted structure of kAE1 showing amino acids mutated in each kAE1 mutant. kAE1 R295H is a recessively inherited substitution in the N-terminal cytosolic domain of the protein. kAE1 Y413H is a dominantly inherited substitution in transmembrane domain (TM) 1 of the core domain. In the gate domain, S525F and R589H substitutions occur in TM 5 and TM 6, respectively. We generated the newly identified dRTA variant cDNAs and expressed them or kAE1 WT in mIMCD3 cells to assess protein abundance (**Figure 1B**), localization (**Figure 1C, D**), transport activity (**Figure 1E**) and lifespan (**Figure 1F**). As seen on **Figure 1B**, both kAE1 R295H and S525F variants display 2 typical bands similar to kAE1 WT. The top band encompasses kAE1 proteins carrying a complex oligosaccharide that have reached the Golgi and beyond, while the bottom band corresponds to high mannose-carrying kAE1 proteins located in the endoplasmic reticulum. However, kAE1 Y413H mutant bands intensity was overall weaker than WT and displayed a predominant single band aligned with high mannose-carrying kAE1 proteins. These results indicate that the 3 newly described dRTA mutants are expressed in mIMCD3 cells. We next localized these mutants by immunofluorescence in mIMCD3 cells. While kAE1 WT is targeted to the plasma membrane, the kAE1 R295H, Y413H and S525F variants partially colocalize with endoplasmic reticulum marker calnexin (**Figure 1C**). Yet, both kAE1 WT and mutants appropriately co-localized with basolateral membrane marker beta-catenin in polarized mIMCD3 cells (**Figure 1D**). However, when the transport function was assayed on confluent cells, both kAE1 Y413H and S525F had significantly lower transport activity than WT (**Figure 1E**) and were not significantly different from un-induced WT-Dox cells (not shown on the graph). Finally, we observed that degradation of the kAE1 Y413H mutant begins 6 hours post-synthesis while kAE1 WT abundance remained stable for 24 hours (**Figure 1F**). Given this premature degradation, we did not perform further assays on cells expressing the Y413H protein. Overall, these results indicate that except for kAE1 Y413H mutant, the other mutants are expressed and traffic to the basolateral membrane of polarized mIMCD3 cells, similar to the previously published kAE1 R589H mutant.

Expression of the kAE1 R295H, S525F and R589H dRTA mutants alter autophagy processes in mIMCD3 cells and whole kidney lysates

In mice expressing kAE1 R607H, a striking reduction in type-A intercalated cells was noted and in the remaining cells, autophagy marker p62 and ubiquitin accumulated in these abnormally enlarged cells¹⁸. We therefore investigated the autophagy machinery in dRTA mutant mIMCD3 cells and in R607H knockin (KI) mice. We first examined the ratio of autophagosome marker LC3BII protein relative to the total intensities of LC3BI and LC3BII as well as p62 levels. These experiments were performed at steady state, upon autophagy induction by starvation, or after autophagy inhibition by Bafilomycin (Baf) A1 (**Figure 2A-C**). **Figure 2A – I** show a consistent increase in the ratio of LC3B II to total LC3B (I + II) in the mutant cells at steady state (**panel D**), with starvation (**panel F**) and with Baf A1 (**panel H**) except for the kAE1 R295H mutant which was not significantly different from WT at steady state only. This suggests an altered autophagy process in the mutant cells, in agreement with preliminary findings from Mumtaz and colleagues¹⁸. There was no significant difference in p62 abundance in mutant cells compared to WT at steady state and with starvation (**Figure 2E-G**). However, with Baf A1, R589H mutant cells had significantly lower p62 abundance compared to WT (**Figure 2I**). To confirm these findings, we

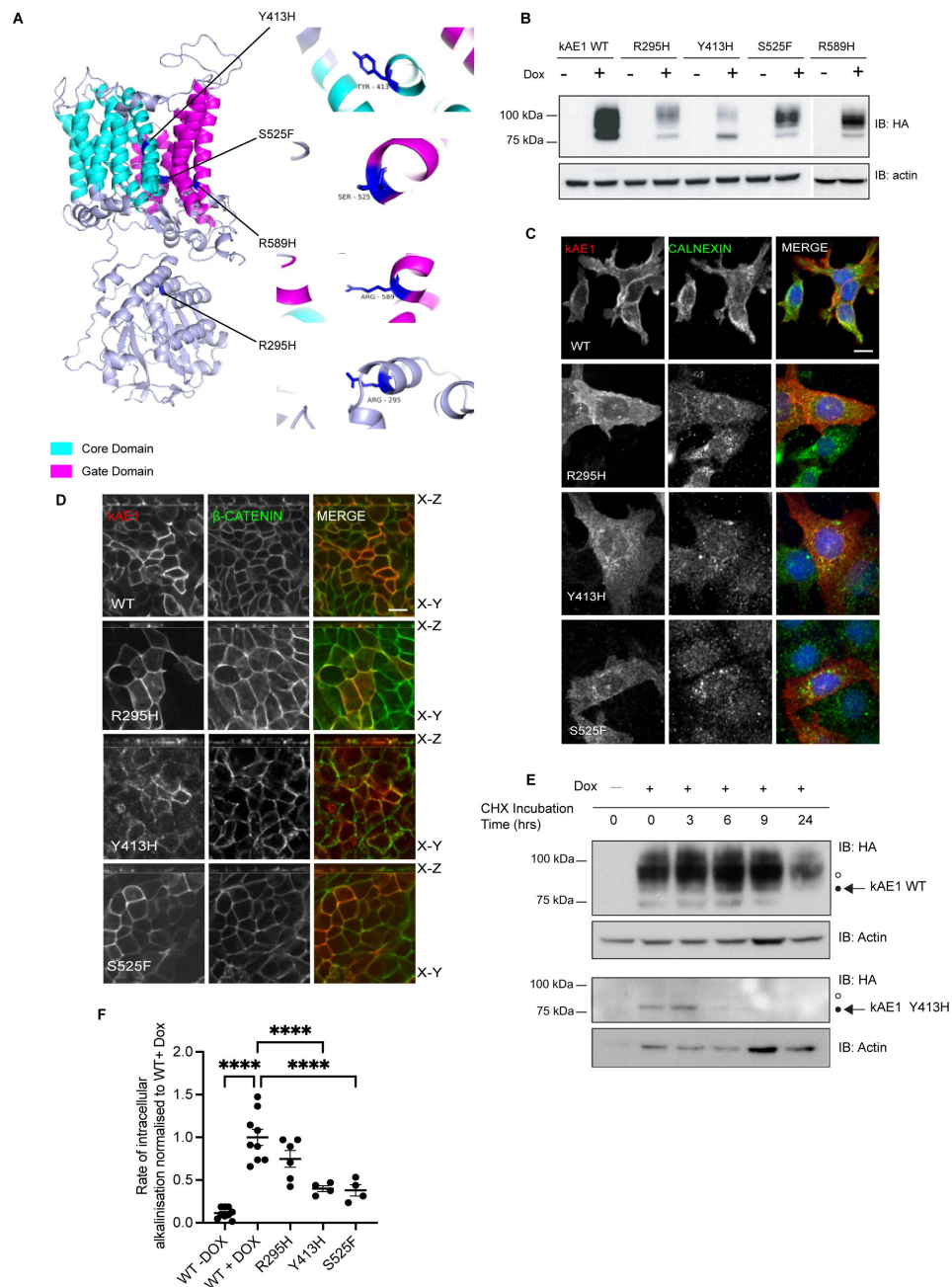


Figure 1.

kAE1 R295H, Y413H, S525F and R589H dRTA mutants are either dysfunctional or prematurely degraded.

(A) Alpha Fold predicted structure of the kidney isoform of Band 3 anion exchanger 1 (kAE1) with core and gate domains highlighted. The dRTA kAE1 mutation sites are coloured in blue with line extensions detailing specific amino acids mutated. (B) Immunoblot showing kAE1 expression and corresponding actin band of mIMCD kAE1 WT, R295H, Y413H, S525F and R589H cells treated with and without doxycycline for 24 hrs. Mouse anti-HA antibody was used to detect kAE1-HA, top and bottom bands correspond to kAE1 carrying complex and high mannose oligosaccharides, respectively. Immunostaining of kAE1 WT or mutant (red) and calnexin (green) in non-polarized mIMCD3 cells (C), and kAE1 WT or mutant (red) and b-catenin (green) in polarized (D) mIMCD cells. Scale bar = 10 μ m. Red = kAE1, Green = β -Catenin. (E) Rate of intracellular alkalinisation in WT or mutant mIMCD3 cells normalized to WT+ Dox. **** indicates $P < 0.0001$ using one-way ANOVA followed by a Dunnett's post-hoc test. Error bars correspond to mean $\pm$ SEM, n = minimum 4. (F) Immunoblot of cycloheximide (CHX) chase assay with corresponding actin in kAE1 mIMCD WT and Y413H cells showing the degradation of kAE1 Y413H after 3hrs CHX incubation.

next assessed the abundance of these markers in whole kidney lysates from the kAE1 R607H KI mice. We observed that the total LC3B abundance was significantly higher in homozygous KI mice (KI/KI) compared to WT, with no difference in p62 (**Figure 2J-L**). Overall, these results suggest an abnormal autophagy in mutant mIMCD3 cells and in kidneys of R607H KI mice.

dRTA kAE1 mutant expressing cells have more alkaline intracellular pH than WT and altered autophagy flux

Given these preliminary findings of abnormal autophagy in the dRTA mutant cells, we next assessed the efficiency of the autophagy machinery in WT or dRTA mutant-expressing cells using the eGFP-RFP-LC3 construct (**Figure 3A**). The green fluorescent protein fused to LC3 is quenched in the acidic environment of the autolysosome, while both eGFP and RFP fluoresce in vesicles in the neutral lumen of the autophagosome^{27,28}. Focusing on cells expressing either kAE1 S525F or R589H, we quantified the number of eGFP+ (green), RFP+ (red) and double eGFP+/RFP+ (yellow) vesicles in the kAE1 mutant or WT transfected live cells (**Figure 3B-D**). kAE1 S525F mutant cells have significantly more autophagosomes (yellow) than WT counterparts (**Figure 3B**), and both kAE1 S525F and R589H have significantly more autolysosomes (red) than WT (**Figure 3C**).

Figure 3D shows that both mutants, particularly the kAE1 S525F mutant cells had significantly more autophagosomes than autolysosomes. This finding suggests an upregulation of autophagy and inhibition in the downstream steps of autophagy that involves the fusion of the autophagosome with the lysosome²⁹. As autolysosomes require luminal v-H⁺-ATPase -dependent acidification to efficiently clear cell debris³⁰ and as kAE1 is a base transporter whose function is linked to the v-H⁺-ATPase, we next wondered whether the steady-state cytosolic pH of mIMCD3 cells expressing kAE1 WT or mutant was different. Using BCECF-AM, we observed that the steady state intracellular pH (pHi) of kAE1 mutant cells was more alkaline than WT cells (except for kAE1 R295H cell pHi which has a similar trend but is not significantly different from WT) (**Figure 3E**), in agreement with transport activity (**Figure 1F**). Given the lack of difference in phenotype between the kAE1 R295H and kAE1 WT mIMCD3 cells, we focused the subsequent experiments on kAE1 S525F and R589H mutant cells. As a higher intracellular pH could contribute to alterations in the autophagy machinery³¹, we chemically acidified the cytosolic pH of cells expressing kAE1 mutant or WT and examined autophagy and lysosomal markers (**Figure 3 F-H**). We first determined that incubation of mIMCD3 cells in 0.033 μ M nigericin for 2 hours acidified cytosolic pH to 6.9 without causing cell death (Supplementary Figure 1). We observed that chemically reducing pHi to 6.9 in mutant expressing cells reduced the ratio of LC3B II to total LC3B (**Figure 3I**) and the abundance of lysosomal-associated membrane protein 1 (LAMP1) in R589H cells (**Figure 3 J**) to levels similar to WT cells at steady state. These findings suggest that abnormal autophagy in the mutant cells may be caused by their alkaline pHi, resulting from a reduced anion exchange activity of the mutant kAE1 protein.

mIMCD3 cells expressing dRTA kAE1 mutants and R607H KI kidney tissues have abnormal lysosome number and size

The accumulation of autophagosomes and autolysosomes as seen above may suggest one or a combination of the following: an inability of autophagosomes to fuse with lysosomes and/ or a defect in lysosomal degradative activity in the mutant cells^{19,32}. We first examined the lysosomal degradative activity by assessing lysosomal protease Cathepsin B activity using Magic Red staining (**Figure 4A-B**). In agreement with increased RFP+ vesicles (**Figure 3D**), the kAE1 S525F mutant cells had a significantly higher number of Magic Red positive vesicles than WT whereas the kAE1 R589H mutant cells had significantly larger Magic Red positive vesicles, suggesting an accumulation of undigested material³³. To validate this finding, we performed immunostaining and quantified LAMP1 positive staining in β 1 ATPase positive cells (a marker of ICs) in WT and R607H KI mouse kidney sections. The KI mice showed significantly more and larger LAMP1 positive vesicles compared to WT mice in both cortex and medulla (**Figure 4 C-F**). We

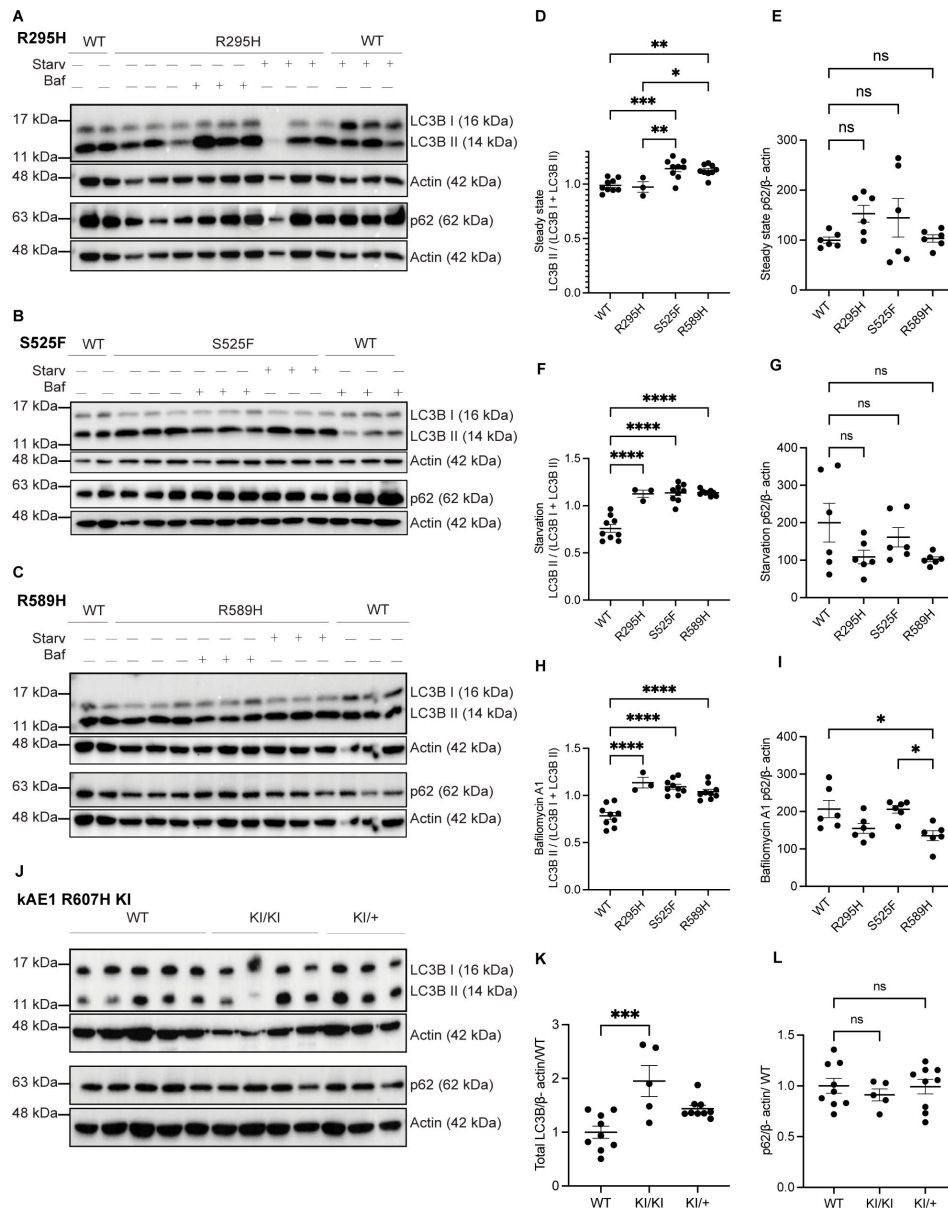


Figure 2.

Autophagy is upregulated in dRTA KAE1 mutants *in vitro* and *in vivo*.

(A-C) Immunoblots of LC3 B and p62 with corresponding actin abundance in KAE1 WT, R295H, S525F and R589H mIMCD3 cells at steady state, under starvation (Starv) or 400 nM Bafilomycin A1 (Baf) treatment. (D-I) Quantification of immunoblots shown in A-C showing the ratio of LC3B II to total LC3B and p62. Error bars correspond to mean $\pm$ SEM, n= minimum 3. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.005$, **** $P < 0.001$ using one-way ANOVA followed by a Tukey's post-hoc test. Immunoblots (J) and quantification (K, L) of LC3B and p62 abundance in KAE1 R607H KI mouse whole kidney lysates Error bars correspond to mean $\pm$ SEM, n= minimum 5. *** $P < 0.005$ using one-way ANOVA followed by a Tukey's post-hoc test.

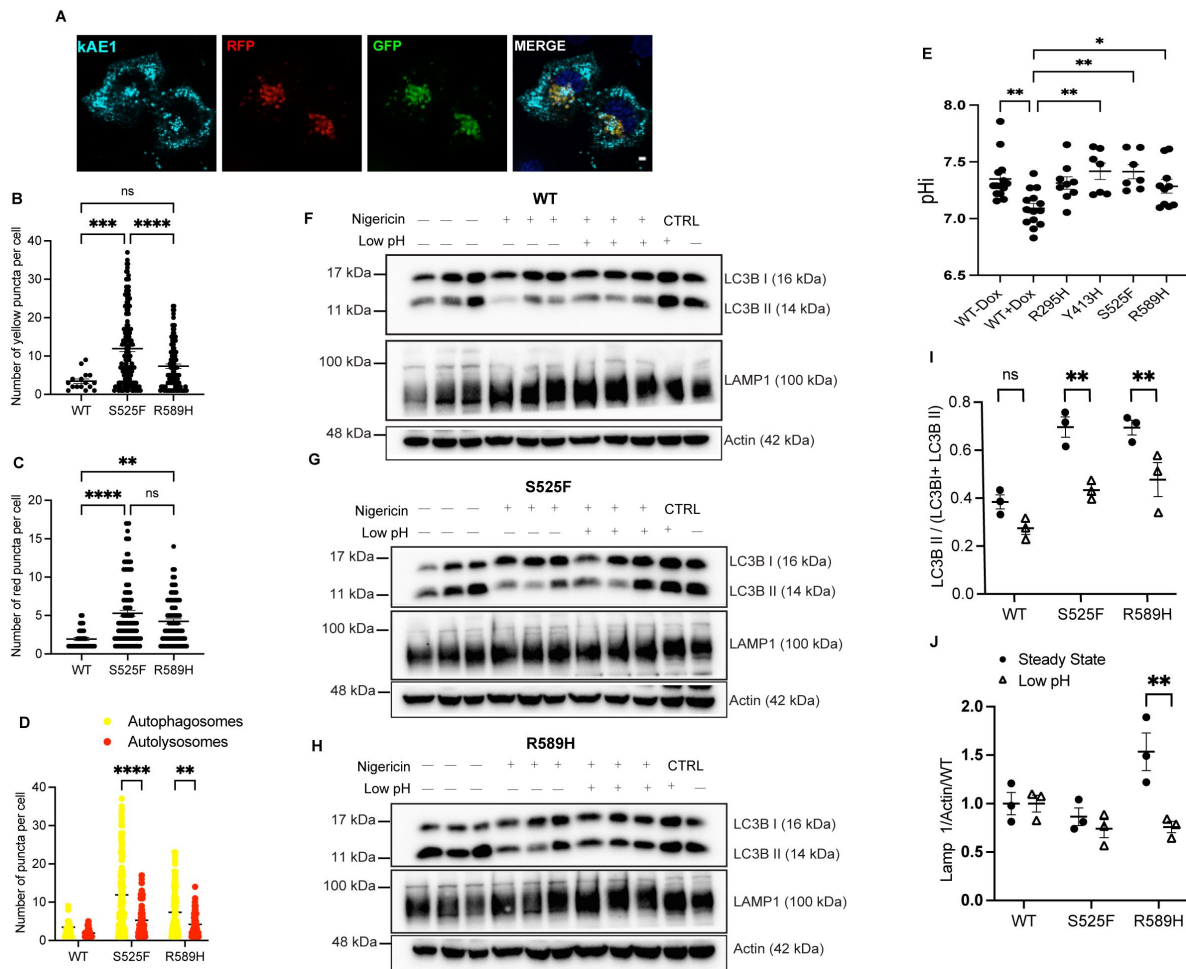


Figure 3.

DRTA KAE1 mutants have more alkaline steady-state intracellular pH and altered autophagy flux.

(A) Immunofluorescence staining of kAE1 in eGFP-RFP-LC3 transfected mIMCD3 cells expressing kAE1. GFP = green, RFP = red, kAE1 = cyan. Scale bar = 8 μ m. Graphical representation of number of yellow (autophagosomes) (B) and red (autolysosomes) (C) puncta per cell expressing kAE1. Error bars correspond to mean $\pm$ SEM, n = minimum 32. ** $P < 0.01$, *** $P < 0.005$, **** $P < 0.001$ using one-way ANOVA followed by a Tukey's post-hoc test. (D) Grouped graph of the number of yellow (autophagosomes) and red (autolysosomes) puncta per cell expressing kAE1 respectively. Note that the statistical analysis displayed only compared yellow and red groups for simplification. Error bars correspond to mean $\pm$ SEM, n = minimum 32. ** $P < 0.01$, **** $P < 0.001$ using two way ANOVA followed by a Sidak's post-hoc test. (E) Graphical representation of intracellular pH measurement of mIMCD kAE1 WT, R295H, Y413H, S525F and R589H cells. Error bars correspond to mean $\pm$ SEM, n = minimum 32. * $P < 0.05$, ** $P < 0.01$ using one-way ANOVA followed by a Dunnett's post-hoc test. (F-H) Immunoblot of LC3B, LAMP1 and actin in kAE1 WT, S525F and R589H mIMCD3 cells at steady state and under chemically reduced intracellular pH conditions. Graphical representation of the ratio of LC3B II to total LC3B ratio (I) or LAMP 1 (J) at steady state versus at low pHi in mIMCD3 kAE1 WT, S525F and R589H. Black circles indicate steady state cells and triangles indicate low pHi cells. Error bars correspond to mean $\pm$ SEM, n = 3. ** indicates $P < 0.01$ using two-way ANOVA followed by a Sidak's post-hoc test.

probed further into the lysosomal activity by quantifying lysosomal protease Cathepsin D (immature, intermediate and mature) protein abundance by immunoblot in isolated primary murine A-ICs. Although the abundance of immature and intermediate cathepsin D did not differ between genotypes, the KI mice showed a significantly decreased abundance of mature cathepsin D (Supplementary Figure 2). Thus, in line with *in vitro* findings, A-IC from homozygous R607H KI mice display relatively more and larger lysosomes with reduced active protease abundance than WT littermates, suggesting a lysosomal defect in the dRTA kAE1 mutant cells.

dRTA kAE1 mutant cells have lower ATP production rate and abnormal mitochondrial content

Lysosomal degradation is highly dependent on a low luminal pH generated in part by the vacuolar-type H⁺-ATPase³⁴ whose activity depends on ATP hydrolysis. We therefore analysed the ATP production rate in mIMCD3 cells, specifically glycolysis and oxidative phosphorylation. We measured the oxygen consumption rate (OCR) (Figure 5A) and extracellular acidification rate (ECAR) in empty vector transfected, kAE1 WT or mutant cells (Figure 5B). Both kAE1 S525F and R589H mutant cells had a lower ATP production rate compared to WT (Figure 5C). More specifically, the R589H mutant cells had a lower mitochondrial ATP production rate (Figure 5D) whereas the kAE1 S525F mutant cells exhibited a lower glycolytic ATP production rate (Figure 5E). With the mitochondria being the major ATP-producing organelles³⁵, we assessed mitochondrial content by immunostaining of translocase of the outer membrane 20 (TOM20) both *in vitro* and *in vivo*. Both kAE1 S525F and R589H mutant cells have higher mitochondrial content compared to WT as determined by total overall intensity of TOM20 positive puncta (Figure 5F&G). In line with this result, although a decreased fluorescence intensity was observed in the cortex, there was a significantly higher TOM20 fluorescence intensity in medullary kidneys of homozygous R607H KI mutant mice compared to WT littermates (Figure 5H-K).

Discussion

In this study, we characterized three newly identified dRTA-causing kAE1 variations. Combining *in vivo* and *in vitro* studies, we demonstrated that reduced transport activity of the kAE1 mutants correlated with increased cytosolic pH, reduced ATP synthesis, attenuated downstream autophagic pathways and lysosomal dysfunction, pertaining to the fusion of autophagosomes and lysosomes and/ or lysosomal degradative activity.

In line with previous observations¹⁸, the kAE1 R295H, Y413H and S525F mutants were intracellularly retained in non-polarised mIMCD3 cells but properly localized to the basolateral membrane after polarization. When expressed in Madin-Darby canine kidney (MDCK) cells, other dRTA-causing kAE1 mutants such as dRTA R602H, G701D, V488M, deltaV850 variants also exhibited a plasma membrane trafficking defect^{10,36,37}. In contrast, the kAE1 R589H mutant was correctly targeted to the plasma membrane¹⁸. Functionally, cells expressing the Y413H and S525F mutants exhibit about 40 % reduction in chloride/bicarbonate exchange activity compared to kAE1 WT, similar to the previously characterized recessive G701D mutant but unlike the R295H mutant³⁸. Therefore, the mechanism causing dRTA remains unclear in the case of the newly identified R295H variant. These findings add to the growing list of *SLC4A1* gene variations causing dRTA.

We next investigated the roots of the altered autophagy briefly reported in R607H KI mice¹⁸ using mIMCD3 cells and whole kidney lysates. In kidneys of the KI mice, a decrease in A-IC, accumulation of p62 and ubiquitin and enlarged remaining A-ICs, prompted us to investigate autophagy pathways in dRTA mutant cells and in homozygous R607H KI mice^{18,38}. During the autophagy process, LC3B I (a marker for autophagosomes) is converted to lipidated LC3B II³⁹.

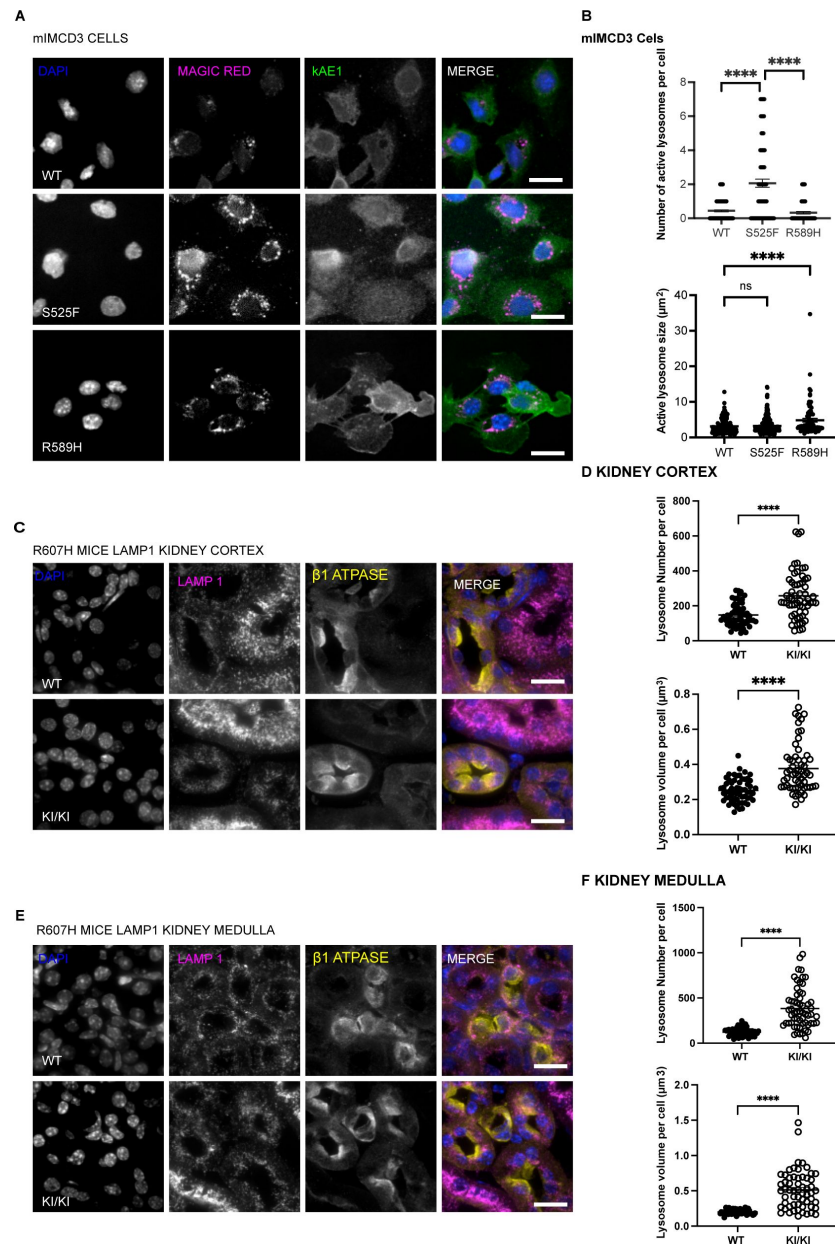


Figure 4.

dRTA KAE1 mutants have bigger or more lysosomes than WT *in vitro* and *in vivo*.

(A) Immunofluorescence images of KAE1 WT, S525F and R589H mIMCD3 cells incubated with Magic Red substrate for 1hr at 37°C in the dark. Green = KAE1, magenta = active lysosomes, blue = nuclei. Scale bar = 16 μm . (B) Graphical representation of number and size of active lysosomes per cell. Error bars correspond to mean $\pm$ SEM, n= minimum 30. **** P<0.0001 using one-way ANOVA followed by Tukey's post-hoc test. Immunofluorescence images of LAMP 1 and β 1 ATPase in kidney cortex (C) or medulla (E) from KAE1 WT and R607H KI mice. Blue = nuclei, magenta = LAMP 1 (lysosomes), Yellow = β 1 ATPase. Scale bar = 8 μm . Graphical representation of the number and volume of LAMP1 vesicles in β 1 ATPase positive cells in the kidney cortex (D) or medulla (F) of WT or R607H KI mice. Error bars correspond to mean $\pm$ SEM, n= 60. **** P<0.001 using Student's t-test.

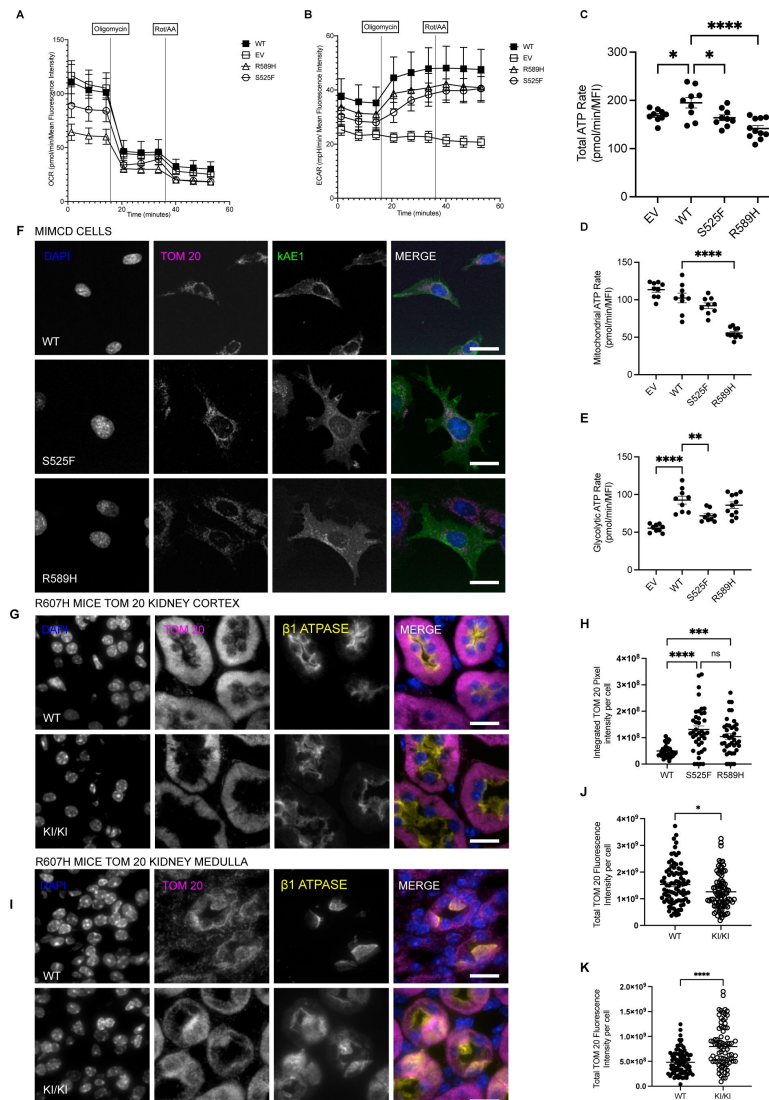


Figure 5.

dRTA KAE1 mutant cells have lower ATP production rate and abnormal mitochondrial content.

(A - B) Oxygen consumption rate (OCR) and Extra Cellular Acidification Rate (ECAR) of empty vector transfected cells, kAE1 WT, S525F or R589H mIMCD3 cells analyzed in a Seahorse XFe96 Extracellular Flux Analyzer with the ATP Rate Assay Test Kit. The empty vector transfected cells provided a control for a potential effect of doxycycline on measurements. (C) Graphical representation of the combination of ATP production rates from mitochondrial respiration (mitoATP) and glycolysis (glycoATP) of kAE1 WT, S525F and R589H mIMCD3 cells measured in real-time following sequential injections of oligomycin and Rotenone +Antimycin A. Error bars correspond to mean $\pm$ SEM, n= minimum. * $p < 0.05$, **** $p < 0.0001$ using one-way ANOVA followed by Tukey's post-hoc test. Graphical representations of mitochondrial respiration (D) and glycolytic ATP production (E) in kAE1 WT, S525F and R589H mIMCD3 cells. Error bars correspond to mean $\pm$ SEM, n= minimum 8. *** $p < 0.01$, **** $p < 0.0001$ using one-way ANOVA followed by Tukey's post-hoc test. (F) Immunofluorescence staining of TOM 20 and kAE1 in kAE1 WT, S525F and R589H mIMCD3 cells. Blue = nuclei, Magenta = TOM 20, Green = kAE1. Scale bar = 8 μ m. (G) Graphical representation of total TOM 20 fluorescence intensity per cell expressing kAE1. Error bars correspond to mean $\pm$ SEM, n= minimum 40. *** $p < 0.001$, **** $p < 0.0001$ using one-way ANOVA followed by Tukey's post-hoc test. Immunofluorescence images of TOM 20 and β 1 ATPase in kidney cortex (H) or medulla (I) of kAE1 R607H WT and KI mice exposed to a salt depleted diet with an acid challenge²⁶. Blue = nuclei, magenta = TOM 20 (mitochondria), Yellow = β 1 ATPase. Scale bar = 8 μ m. (J) Graphical representation of the total TOM 20 fluorescence intensity in β 1 ATPase positive cells in the cortex (J) or medulla (K) of the kidney. Error bars correspond to mean $\pm$ SEM, n= 90. * $p < 0.05$, **** $p < 0.0001$ using Student's t-test.

and p62 aggregates to facilitate the degradation of ubiquitinated proteins within the autophagosome complex⁴⁰. In mIMCD3 cells, LC3B lipidation was increased in the kAE1 R295H, S525F and R589H mutant cells compared to WT, an increase that persisted with both Baf A1 treatment and starvation. Although opposite effects were expected under inhibition or induction of autophagy, such similar effect has been previously described. In the proximal tubule of obese mice, LC3B accumulation indicating a stagnated autophagy flux, was observed with both chloroquine treatment and 24 hour starvation⁴¹. Similarly, in NRK-52E cells, a disruption of the autophagy machinery by LC3B II accumulation in high cadmium-stressed cells occurred under either Baf A1 or rapamycin (an autophagy inducer) treatment⁴². Although LC3B II is elevated during both increased autophagy flux and disrupted autophagy, we did not observe significant accumulation of p62 in mIMCD3 cells expressing dRTA mutants. p62 is specifically a marker of autophagy-mediated protein clearance^{43–45}. Therefore, p62 accumulation in R607H KI mouse kidney sections strongly suggests a compromised autophagy-mediated clearance while the increased LC3B lipidation without significant changes in p62 in mIMCD3 cells points towards either an increased autophagic flux and/or a disrupted autophagy.

To obtain a clearer picture of the precise autophagic pathway altered in the dRTA mutants, we probed further into the different stages of autophagy and autophagy flux. We noted an accumulation of autophagosomes and autolysosomes in the S525F and R589H mutant cells. This was recapitulated in the R607H KI mice which showed significantly more and bigger LAMP 1 positive vesicles in both kidney cortex and medulla, suggesting a blockage in autophagy flux in both dRTA mutant cells and KI mice. Such blockage has been implicated in the pathophysiology of several diseases. In lysosomal storage disease, lysosome accumulation in proximal tubule cells is a key component in the pathways mediating epithelial dysfunction¹⁹. In this study, restoring autophagy flux attenuated disease progression. Another study in SK-N-SH, RT4-D6P2T and HeLa cells implicated autophagosome and lysosome accumulation in cellular toxicity associated with neurodegenerative diseases³². In agreement with altered lysosomal function, we also noted a greater abundance and size of active cathepsin B lysosomal protease vesicles in mIMCD3 cells. Increased cathepsin B activity affects lysosomal biogenesis, autophagy initiation and cellular homeostasis⁴⁶. In the renal context, cathepsin B knockout mice demonstrated a higher resistance and quicker recovery from glomerular damage⁴⁷, suggesting that cathepsin B accumulation may be detrimental to the cells. Increased cathepsin B abundance in the dRTA mutant cells also correlates with the accumulation of lysosomes. Overall, these findings suggest that the pathogenesis of dRTA in our models involves an inhibition of autophagy flux at the downstream steps involving autophagosome-lysosome fusion and lysosomal protein clearance^{29,33,34,48,49}.

The question remained as to how these dRTA variants altered autophagy. The *SLC4A1-3* gene family that includes AE1 are regulators of intracellular pH in different cell types^{50–52}. The reduced anion exchange activity in mIMCD3 cells expressing the R589H and S525F variants expectedly correlated with an increased pHi compared to WT counterparts. We wondered whether this was the mechanism explaining blockade in autophagy flux since pHi variations impact autophagy^{31,34,53–55}. Previous studies reported more perinuclear localization of lysosomes and autophagosome-lysosome fusion in cells with an increased pHi^{53,54}. Although not examined in our study, an increase in intracellular pH due to starvation decreased the levels of lysosomal kinesin superfamily member KIF2 and ADP-ribosylation factor-like 8B (ARL8), which are responsible for redistributing lysosomes to the cell periphery. This reduction subsequently inhibited mTORC1, resulting in increased autophagosome synthesis and autophagosome-lysosome fusion⁵³.

While an alkaline cytosolic pH partially explains the impairment in autophagy flux, a close relationship also exists between intracellular pH and cellular energy dynamics and metabolic stress, all of which are key regulators of autophagy⁵⁶. Therefore, it was plausible that kAE1 variant-induced autophagy dysregulation occurs through a signalling pathway akin to that of

energy deprivation-induced autophagy. This hypothesis is supported by our findings of reduced ATP production rate in dRTA kAE1 S525F and R589H mutant cells compared to kAE1 WT cells. We also found that both kAE1 S525F and R589H mutant cells have higher mitochondrial content compared to kAE1 WT cells. The increased mitochondrial content coupled with low ATP point towards improper mitochondrial function in dRTA variant cells. Low ATP levels as seen in dRTA mutant cells may impair autophagy as was shown in human RPE cells⁵⁷. ATP reduction in RPE cells led to complex mitochondrial changes such as structural disorganization, enzyme activity decline, and oxidative damage to mitochondrial components and DNA⁵⁷. Similarly, in pancreatic islet cells, an alkaline pHi led to increased uptake of phosphate by mitochondria, accelerating the production of superoxide, promoting mitochondrial permeability transition, and inducing translational attenuation due to endoplasmic reticulum stress, ultimately impairing insulin secretion⁵⁸. Overall, our data support that expression of the kAE1 variants increases pHi which alters mitochondrial function and leads to reduced cellular energy levels that eventually attenuates energy-dependent autophagic pathways including autophagosome-lysosome fusion and lysosomal protein clearance.

In light of these observations, we postulated that correcting the alkaline pHi of dRTA mutant-expressing mIMCD3 cells will alleviate this blockage in autophagy flux. We observed that a chemically engineered pHi of 6.9⁵⁹ reduced LC3B II accumulation and LAMP1 abundance in mIMCD3 mutant cells to expression levels similar to that of WT cells at baseline. This suggests that a chemically reduced pHi facilitated protein clearance in the two dRTA mutant cells^{31,53} as noted in other studies. In one such study, treatment of SH-SY5Y cells with FCCP and nigericin also acidified intracellular pH and triggered autophagy and mitophagy³¹. Similarly, acid loading in proximal tubular cells under chronic metabolic acidosis showed increased autophagic flux and mitophagy⁶⁰. These findings are in line with our results and establish a link between altered autophagy flux and the alkaline pHi of dRTA variant cells.

In conclusion, our study established a strong relationship between the expression of defective kAE1 proteins, reduced mitochondrial activity, decreased autophagy and impaired protein degradative flux. Whether this abnormal degradative pathway explains the premature loss of A-IC will need to be elucidated in further studies.

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Additional files

Supplementary Material and Figures [↗](#)

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Reviewer #1 (Public review):

Summary:

This study is an evaluation of patient variants in the kidney isoform of AE1 linked to distal renal tubular acidosis. Drawing on observations in the mouse kidney, this study extends findings to autophagy pathways in a kidney epithelial cell line.

Strengths:

Experimental data are convincing and nicely done.

Weaknesses:

Some data are lacking or not explained clearly. Mutations are not consistently evaluated throughout the study, which makes it difficult to draw meaningful conclusions.

<https://doi.org/10.7554/eLife.108253.1.sa2>

Reviewer #2 (Public review):

Context and significance:

Distal renal tubular acidosis (dRTA) can be caused by mutations in a Cl/HCO₃⁻ exchanger (kAE1) encoded by the SLC4A1 gene. The precise mechanisms underlying the pathogenesis of the disease due to these mutations are unclear, but it is thought that loss of the renal intercalated cells (ICs) that express kAE1 and/or aberrant autophagy pathway function in the remaining ICs may contribute to the disease. Understanding how mutations in SLC4A1 affect cell physiology and cells within the kidney, a major goal of this study, is an important first step to unraveling the pathophysiology of this complex heritable kidney disease.

Summary:

The authors identify a number of new mutations in the SLC4A1 gene in patients with diagnosed dRTA that they use for heterologous experiments in vitro. They also use a dRTA mouse model with a different SLC4A1 mutation for experiments in mouse kidneys. Contrary to previous work that speculated dRTA was caused mainly by trafficking defects of kAE1, the authors observe that their new mutants (with the exception of Y413H, which they only use in Figure 1) traffic and localize at least partly to the basolateral membrane of polarized heterologous mIMCD3 cells, an immortalized murine collecting duct cell line. They go on to show that the remaining mutants induce abnormalities in the expression of autophagy markers and increased numbers of autophagosomes, along with an alkalinized intracellular pH. They also reported that cells expressing the mutated kAE1 had increased mitochondrial content coupled with lower rates of ATP synthesis. The authors also observed a partial rescue of the effects of kAE1 variants through artificially acidifying the intracellular pH. Taken together, this suggests a mechanism for dRTA independent of impaired kAE1 trafficking and dependent on intracellular pH changes that future studies should explore.

Strengths:

The authors corroborate their findings in cell culture with a well-characterized dRTA KI mouse and provide convincing quantification of their images from the in vitro and mouse experiments.

Weaknesses:

The data largely support the claims as stated, with some minor suggestions for improving the clarity of the work. Some of the mutants induce different strengths of effects on autophagy and the various assays than others, and it is not clear why this is from the present manuscript, given that they propose pH_i and the unifying mechanism.

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Reviewer #3 (Public review):

Summary:

The authors have identified novel dRTA causing SLC4A1 mutations and studied the resulting kAE1 proteins to determine how they cause dRTA. Based on a previous study on mice expressing the dRTA kAE1 R607H variant, the authors hypothesize that kAE1 variants cause an increase in intracellular pH, which disrupts autophagic and degradative flux pathways. The authors clone these new kAE1 variants and study their transport function and subcellular localization in mIMCD cells. The authors show increased abundance of LC3B II in mIMCD cells expressing some of the kAE1 variants, as well as reduced autophagic flux using eGFP-RFP-LC3. These data, as well as the abundance of autophagosomes, serve as the key evidence that these kAE1 mutants disrupt autophagy. Furthermore, the authors demonstrate that decreasing the intracellular pH abrogates the expression of LC3B II in mIMCD cells expressing mutant SLC4A1. Lastly, the authors argue that mitochondrial function, and specifically ATP synthesis, is suppressed in mIMCD cells expressing dRTA variants and that mitochondria are less abundant in AICs from the kidney of R607H kAE1 mice. While the manuscript does reveal some interesting new results about novel dRTA causing kAE1 mutations, the quality of the data to support the hypothesis that these mutations cause a reduction in autophagic flux can be improved. In particular, the precise method of how the western blots and the immunofluorescence data were quantified, with included controls, would enhance the quality of the data and offer more supportive evidence of the authors' conclusions.

Strengths:

The authors cloned novel dRTA causing kAE1 mutants into expression vectors to study the subcellular localization and transport properties of the variants. The immunofluorescence images are generally of high quality, and the authors do well to include multiple samples for all of their western blots.

Weaknesses:

Inconsistent results are reported for some of the variants. For example, R295H causes intracellular alkalinization but also has no effect on intracellular pH when measured by BCECF. The authors also appear to have performed these in vitro studies on mIMCD cells that were not polarized, and therefore, the localization of kAE1 to the basolateral membrane seems unlikely, based upon images included in the manuscript. Additionally, there is no in vivo work to demonstrate that these kAE1 variants alter intracellular pH, including the R607H mouse, which is available to the authors. The western blots are of varying quality, and it is often unclear which of the bands are being quantified. For example, LAMP1 is reported at 100kDa, the authors show three bands, and it is unclear which one(s) are used to quantify protein abundance. Strikingly, the authors report a nonsensical value for their quantification of LCRB II in Figure 2, where the ratio of LCRB II to total LCRB (I + II) is greater than one. The control experiments with starvation and bafilomycin are not supportive and significantly reduce enthusiasm for the authors' findings regarding autophagy. There are labeling errors

between the manuscript and the figures, which suggest a lack of vigilance in the drafting process.

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